ConvFormerSR: Fusing Transformers and Convolutional Neural Networks for Cross-Sensor Remote Sensing Imagery Super-Resolution

Junjie Li, Yizhuo Meng, Chongxin Tao, Zhen Zhang, Xining Yang, Zhe Wang, Xi Wang, Linyi Li, and Wen Zhang

***Abstract*— Super-resolution (SR) techniques based on deep learning have a pivotal role in improving the spatial resolution of images. However, remote sensing images exhibit ground objects characterized by diverse types, intricate structures, substantial size discrepancies, and noise. Simultaneously, variations in imaging mechanisms, imaging time, and atmospheric conditions among different sensors result in disparities in image quality and surface radiation. These factors collectively pose challenges for existing SR models to fulfill the demands of the domain. To address these challenges, we propose a novel cross-sensor SR framework (ConvFormerSR) that integrates transformers and convolutional neural networks (CNNs), catering to the heterogeneous and complex ground features in remote sensing images. Our model leverages an enhanced transformer structure to capture long-range dependencies and high-order spatial interactions, while CNNs facilitate local detail extraction and enhance model robustness. Furthermore, as a bridge between the two branches, a feature fusion module (FFM) is devised to efficiently fuse global and local information at various levels. Additionally, we introduce a spectral loss based on the remote sensing ratio index to mitigate domain shift induced by cross-sensors. The proposed method is validated on two datasets and compared against existing state-of-the-art SR models. The results show that our proposed method can effectively improve the spatial resolution of Landsat-8 images, and the model performance is significantly better than other methods. Furthermore, the SR results exhibit satisfactory spectral consistency with high-resolution (HR) images.**

***Index Terms*— Convolutional neural networks (CNNs), remote sensing, super-resolution (SR), transformers.**

I. Introduction

LONG-TERM series of remote sensing images, specifically those acquired by Landsat and Sentinel satellites, serve as crucial data sources for a range of applications in the field of earth sciences, including land-use/land-cover change (LUCC) [1], geographic mapping [2], ecosystem services valuation [3], and global climate change monitoring [4]. Higher spatial resolution enables the discernment of finer details and dynamics in the Earth's surface system. However, hardware-related constraints, both technical and cost-related, make it arduous to further improve the spatial resolution of satellite sensors. In light of these challenges, leveraging super-resolution (SR) methods at the software level emerges as a viable approach to enhance the spatial resolution of remote sensing images. SR techniques offer advantages such as cost-effectiveness and efficiency. Consequently, SR has gained substantial attention as a prominent research area in remote sensing [5].

The study of SR techniques originated in the 1960s with the primary objective of recovering and reconstructing a high-resolution (HR) image from a single low-resolution (LR) image [6]. Subsequently, the advent of convolutional neural networks (CNNs) has facilitated significant advancements in CNN-based SR methods. Prominent examples of CNN-based SR networks include SRCNN [7], SRGAN [8], VDSR [9], and EDSR [10]. Deep learning-based SR fundamentally relies on training a substantial number of LR/HR image pairs to establish a mapping model. Additionally, the performance of SR networks can be enhanced through the incorporation of techniques such as residual connections [11], dense connections [12], attention mechanism [13], and generative adversarial networks (GANs) [14]. However, ground objects on remote sensing images have complex structures, substantial size disparities, significant noise, and high local similarity [15]. CNN-based models inherently struggle with modeling long-range dependencies and face limitations in capturing global contextual information due to their constrained receptive field sizes [16]. These aforementioned challenges pose significant obstacles to remote sensing image SR (RSISR).

Recently, deep learning models based on transformers [17] have achieved state-of-the-art performance in various computer vision applications. Transformers demonstrate efficient capability in capturing global contextual information by leveraging self-attention mechanisms. Notably, Liang et al. [16] applied the swin transformer [18] to SR and proposed SwinIR model. Although SR models based on SwinIR and its variants have conducted extensive experiments on natural images, their effectiveness in the context of cross-sensor RSISR necessitates further investigation. Moreover, certain studies have indicated that the performance of a single transformer may not

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necessarily surpass that of CNNs. Because the image patches are squeezed into a 1-D sequence which may lose structural information [19]. Additionally, the application of transformers may disrupt the spectral correlation present in multispectral or hyperspectral remote sensing images [20].

Furthermore, it is important to recognize the substantial distinctions between RSISR and SR tasks for natural images. Presently, supervised learning forms the foundation of most SR approaches in computer vision [5]. However, acquiring paired LR/HR training samples in real-world poses significant challenges. Consequently, the prevailing practice involves downsampling HR images and applying blurring kernels to simulate the degradation process observed in natural images. Nonetheless, this approach is inherently flawed. Remote sensing images with different spatial resolutions taken by different sensors at the same time and same place have provided the real and ideal LR/HR image pairs for the SR model. However, the differences in atmospheric correction, viewing geometry, and bandpass of different sensors introduce domain shift issues in these images [21]. Therefore, it becomes imperative to consider and address the influence of cross-sensor domain shift on RSISR methodologies.

To address the aforementioned challenges, we propose a novel cross-sensor RSISR framework (ConvFormerSR). The framework comprises three key components: shallow feature extraction, deep feature extraction, and upsampling. During the deep feature extraction stage, ConvFormerSR leverages the carefully designed residual attention group (RAG) and residual convolution group (RCG) to capture high-dimensional features at different levels. Subsequently, the feature fusion modules (FFMs) are employed to integrate local details and global contextual information effectively. Moreover, we tackle the issue of domain shift introduced by illumination differences, cloud shadows, terrain changes, and atmospheric conditions through a ratio-based comprehensive spectral loss. This approach ensures spectral consistency between the SR reconstruction results and HR images.

The main contributions of this study are as follows.

1) A global channel attention module (GCAM) is incorporated into the transformer structure, which improves the global modeling capability of the SwinIR limited by the window size.
2) An FFM is proposed to fuse the local details of CNNs and the global contextual information of transformers.
3) A comprehensive spectral loss based on the remote sensing ratio index is constructed to solve the domain shift problem caused by cross-sensors.
4) For the first time, SR is performed simultaneously in four bands: blue, green, red, and near-infrared, and two datasets are constructed to evaluate the effectiveness and performance of the proposed model.

The rest of this article is organized as follows. First, we introduce related work in Section II. Section III describes the methodology of the proposed method in detail. Then, we present the datasets, implementation details, and experimental results in Section IV. Finally, Section V discusses the proposed method, and Section VI concludes our work.

II. Related Work

A. SR Based on CNNs

Dong et al. [7] were the first to introduce CNNs to SR task and proposed the SRCNN model, demonstrating that CNNs can effectively capture the relationship between HR and LR. Subsequently, more and more deep learning-based SR algorithms have been proposed. Shi et al. [22] proposed the well-known sub-pixel (pixel-shuffle) convolution, which generates multichannel feature maps by convolving the LR feature maps, and then reorganizes to obtain HR feature maps. Ledig et al. [8] introduced ResNet [11] and GAN into SR and proposed the SRGAN network. Lim et al. [10] proposed the EDSR network based on SRResNet [8], which mainly improved the residual module, removed the normalization layer, and increased a residual scaling factor to solve the instability problem of the training process. Wang et al. [23] proposed ESRGAN on the basis of SRGAN, replacing the generator with a more complex residual dense network. Zhang et al. [13] introduced the attention mechanism to SR task for the first time. The authors believe that LR images contain a large amount of low-frequency information, which is treated equally between channels and limits the feature expression of CNNs. Niu et al. [24] proposed a holistic attention network (HAN), introducing the layer attention module and channel-spatial attention module to model the holistic interdependencies between convolutional layers, channels, and locations to enhance the feature representation capability of SR networks. Although the above CNNs-based models have achieved good results in SR, they are not effective for long-range dependency modeling [16].

B. SR Based on ViTs

In order to transfer transformers to vision tasks, Dosovitskiy et al. [25] proposed the vision transformers (ViTs), which reshapes the image into a sequence of flattened 2-D patches and then flattens the patches into linear embeddings as the input to the transformer. To reduce the computational complexity of ViT, swin transformer adopts window-based multihead self-attention (W-MSA), and only calculates self-attention inside small windows. Swin transformer has become a new backbone network for many vision tasks. On this basis, Liang et al. [16] proposed SwinIR for image SR, and used swin transformer to extract deep features. Chen et al. [26] proposed a hybrid attention transformer (HAT) based on SwinIR. HAT adds a parallel channel attention block to transformer because they believe that (S)W-MSA captures limited information. In addition, there are some other structures of ViT-based SR models. Yang et al. [27] proposed a novel texture transformer network for image SR, in which the LR and HR images are formulated as queries and keys in a transformer, respectively. Considering the high computational cost of ViT, Lu et al. [28] proposed a novel efficient SR transformer (ESRT) for single image SR. The computation is reduced by a lightweight CNN backbone and a specially designed efficient transformer.

C. Remote Sensing Image SR

The SR methods of remote sensing images are constantly evolving along with the development of natural image SR.

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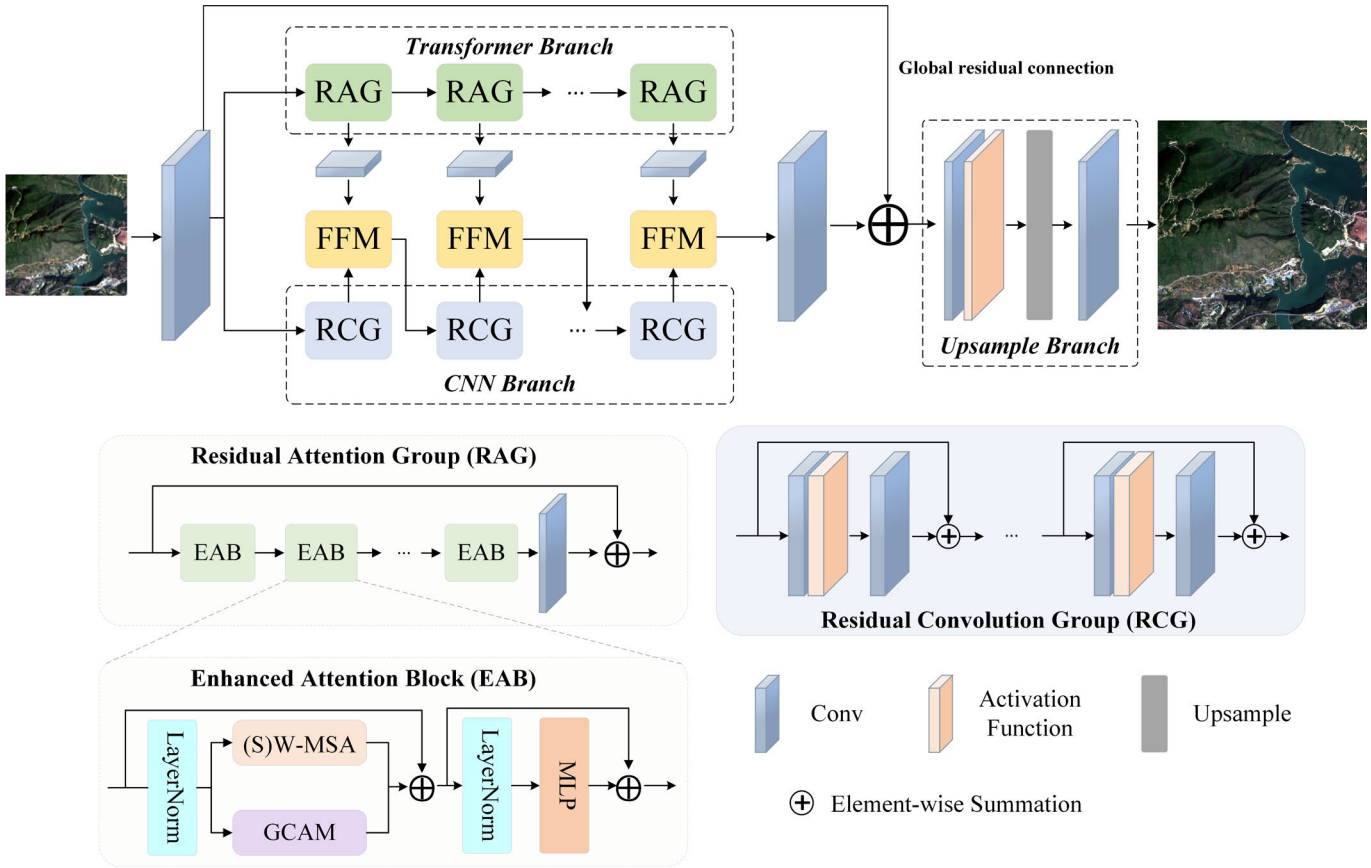


Fig. 1. Architecture of our proposed ConvFormerSR.

After the SRCNN algorithm was proposed, Ducournau and Fablet [29] applied SRCNN to the sea surface temperature (SST) data to achieve SST downscaling. Liebel and Körner [30] applied SRCNN to reconstruct a single Sentinel-2 SR image. Haut et al. [31] proposed a new SR architecture for remote sensing images by combining residual units and skip connections and using 1×1 convolution to achieve reconstruction and reduce the loss of image information. Liu et al. [20] fused transformers and 3-D CNNs to achieve multispectral image SR. Zheng et al. [32] proposed a new unsupervised deep learning-based fusion network (HyCoNet) for hyperspectral and multispectral image fusion. Xu et al. [33] proposed the TE-SAGAN network for RSISR, which mainly designed a self-attention module for extracting global information, introduced a weight normalization layer, and took into account the details and textures of the SR images through various losses such as content loss, perceptual loss, and adversarial loss. Jia et al. [34] proposed a multiattention network (MA-GAN) for RSISR task, which consists of a pyramid convolutional residual dense module, an attention-based upsampling module, and an attention-based fusion module.

For Landsat-8 and Sentinel-2 cross-sensor image SR or image fusion tasks, several scholars have also carried out related research. Shao et al. [35] extended the SRCNN to improve the spatial resolution of Landsat-8 images from 30 to 10 m. On the basis of RCAN, Zhu et al. [36] added channel-spatial attention and dense sampling modules to achieve SR from Landsat-8 LR images to Sentinel-2 HR images. Wu et al. [37] proposed a degradation-term constrained spatio-temporal fusion network that combines residual dense blocks and attention mechanisms to fuse time-series Landsat-8 and Sentinel-2 data to 10-m resolution products. Kong et al. [38] used a dual GAN to enhance the spatial resolution of the Landsat vegetation index. Razzak et al. [39] used the PlanetScope as HR images to further improve the spatial resolution of Sentinel-2 and explore its impact on building detection. CNN-based SR methods have gained prominence in RSISR due to the locality and translational equivariance. However, the application of ViT-based models in RSISR tasks remains uncommon. Moreover, existing RSISR models typically consider the same input channels as natural images. Consequently, these models may not fully exploit the spectral information available in remote sensing images. In addition, existing RSISR models do not consider the domain shift between Landsat-8 and Sentinel-2 data.

III. Method

In this section, we first introduce the overall structure of the proposed ConvFormerSR and then describe three important modules in ConvFormerSR, namely RAG, FFM, and RCG. Finally, we detail our designed loss function.

A. Overall Structure of ConvFormerSR

As illustrated in Fig. 1, ConvFormerSR first obtains the shallow feature $\mathcal{F}$ of LR through a convolutional layer, and

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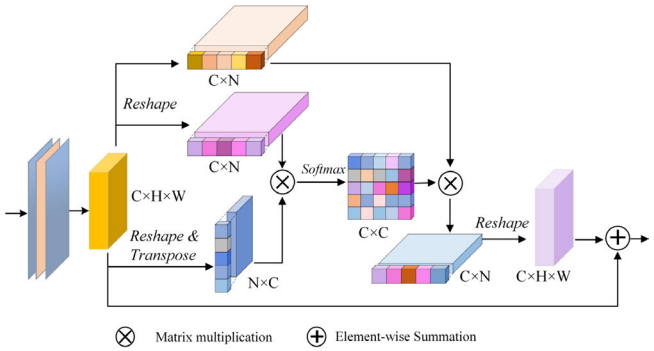


Fig. 2. Structure of GCAM.

then obtains the deep features $\mathcal{D}_T$ and $\mathcal{D}_C$ through the parallel transformer branch and CNN branch, respectively, and fuses the features of the two branches through the FFM module. Subsequently, a global residual connection is used to fuse low-level features and deep features, and finally the pixel-shuffle method [22] is used for upsampling to obtain SR results.

The SR process of ConvFormerSR can be mathematically formulated as

$$I^{\mathrm{SR}} = \mathrm{Up}(\mathrm{G}(\mathcal{D}_T, \mathcal{D}_C) + \mathcal{F}) \tag{1}$$

where $\mathrm{Up}(\cdot)$ represents the upsampling function, and $\mathrm{G}(\cdot)$ represents the feature fusion operation. I^{SR} is the SR reconstructed image.

B. Residual Attention Group

Swin transformer employs a window-based self-attention mechanism to capture the features within the window range, which limits the global representation capability of the model. Furthermore, for pixel-intensive image reconstruction tasks, transformers treat each pixel as a token, which leads to a limited spatial size for image input, and more features are contained in rich channel dimensions. Additionally, previous studies have indicated that convolutional layers can assist transformers in optimizing parameters [26]. To address these limitations, we propose the integration of a GCAM into the standard swin transformer layer (STL) [16]. This enhancement aims to fully leverage the feature representation ability of convolutional layers while harnessing the global modeling capacity of swin transformer.

As shown in Fig. 2, GCAM contains two convolutional layers with an activation function and a channel attention module from DANet [40]. Given a feature map $A \in \mathbb{R}^{C\times H\times W}$, the feature map $B \in \mathbb{R}^{C\times H\times W}$ is obtained after two convolutional layers. It can be expressed as

$$B = W_1\sigma(W_0(A)) \tag{2}$$

where σ represents the GELU activation function, $W_0 \in \mathbb{R}^{(C/\alpha)\times C}$ and $W_1 \in \mathbb{R}^{C\times(C/\alpha)}$. α is a channel reduction ratio. By first compressing and then expanding the channel dimension, the model can effectively learn the nonlinear interaction relationships between different channels while constraining the model complexity [41].

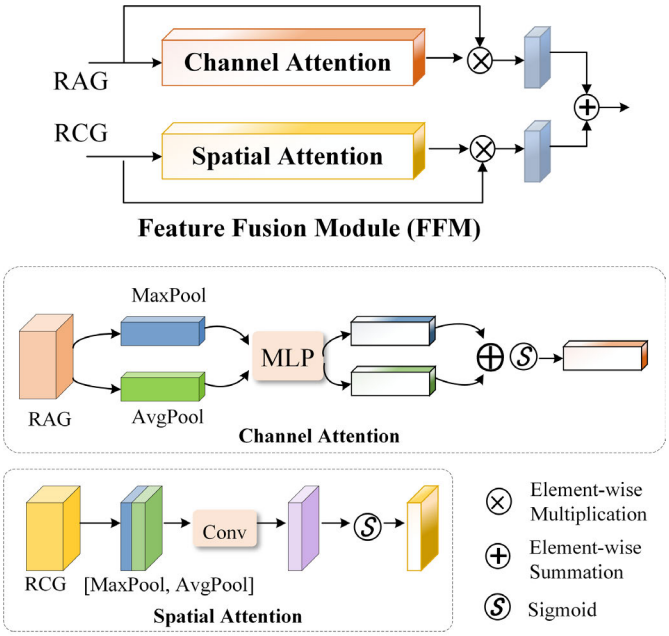


Fig. 3. Structure of FFM.

Subsequently, important channels are selectively emphasized by integrating correlated features among all channels. The calculation of GCAM is as follows:

$$\begin{aligned} C &= f(\mathrm{B}), \quad C \in \mathbb{R}^{C\times N} \\ D &= f(\mathrm{B})^T, \quad D \in \mathbb{R}^{N\times C} \\ E &= \mathrm{softmax}(C \otimes D) \otimes f(\mathrm{B}), \quad E \in \mathbb{R}^{C\times N} \\ \mathrm{GCAM} &= B + \beta f(\mathrm{E}), \quad \mathrm{GCAM} \in \mathbb{R}^{C\times H\times W} \end{aligned} \tag{3}$$

where C, D, and E represent feature maps, $f(\cdot)$ is reshape operation and $\otimes$ represents matrix multiplication. β is a scale parameter that is automatically learned and optimized from 0.

At this point, the enhanced attention block (EAB) can be expressed as

$$\begin{aligned} \hat{s}^l &= (S)\text{W-MSA}\left(\mathrm{LN}\left(s^{l-1}\right)\right) + \gamma\mathrm{GCAM}\left(\mathrm{LN}\left(s^{l-1}\right)\right) + s^{l-1} \\ s^l &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{s}^l\right)\right) + s^l \end{aligned} \tag{4}$$

where s^{l-1} and s^l represent the $(l-1)$th and lth EAB feature maps, respectively. (S)W-MSA is a multihead self-attention module proposed by swin transformer, including window-based multihead self-attention (W-MSA) and shifted window-based self-attention (SW-MSA). (S)W-MSA is formulated as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}} + b\right)V \tag{5}$$

where Q, K, and V represent query, key, and value matrices, respectively. d represents the dimension of query and key, and b is a relative position bias. Like previous work [16], [26], W-MSA and SW-MSA are used alternately. LN and MLP represent LayerNorm layer and multilayer perceptron, respectively. In addition, the parameter γ is used to stabilize the model training. Finally, RAG can be formulated as

$$\mathrm{RAG}^k = \mathrm{RAG}^{k-1} + Ws^l\left(s^{l-1}\left(\ldots s^1\left(\mathrm{RAG}^{k-1}\right)\ldots\right)\right) \tag{6}$$

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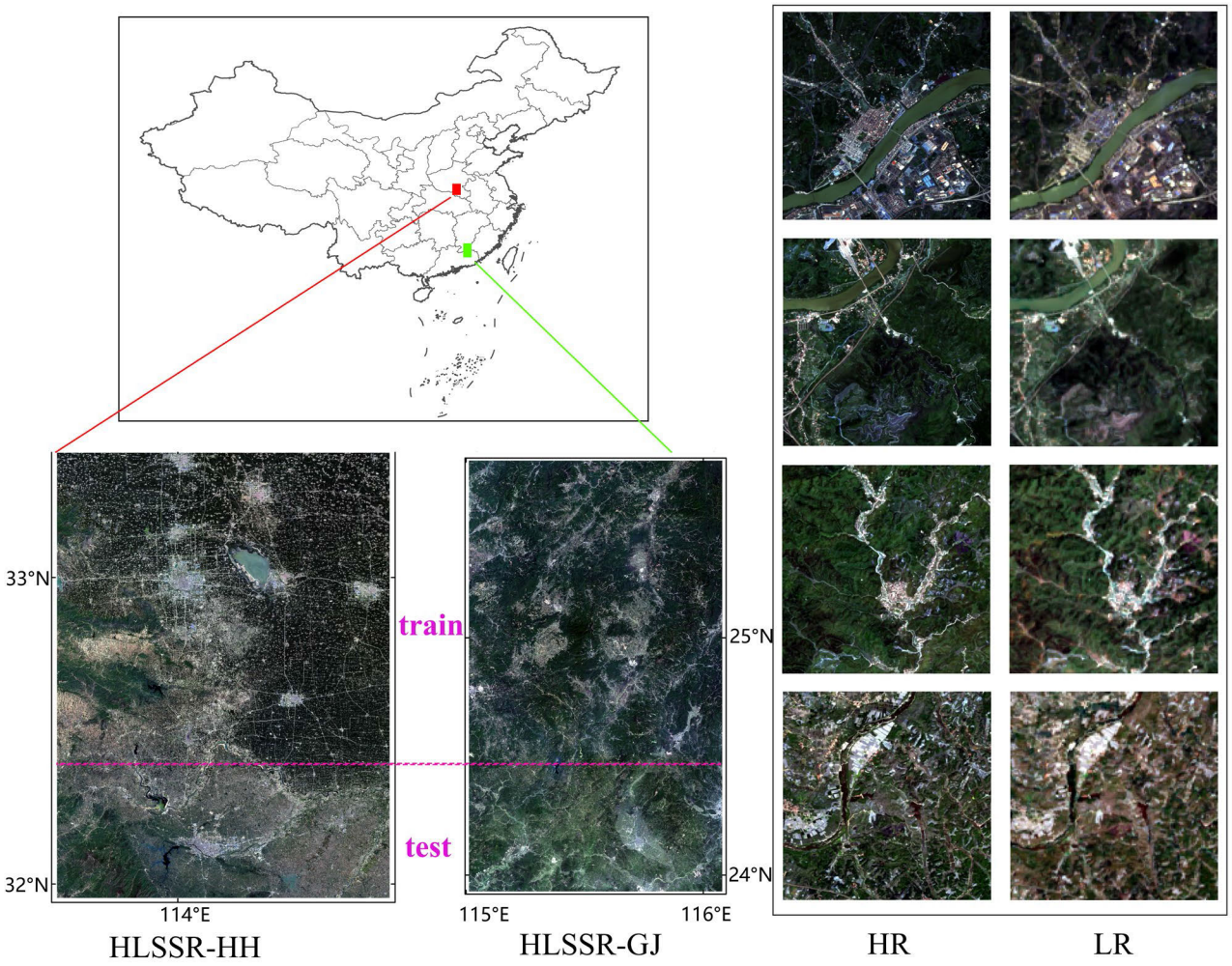


Fig. 4. Geographic locations and HR/LR sample examples of the two datasets.

TABLE I
DETAILS OF HLSSR-HH AND HLSSR-GJ DATASETS

	HLSSR-HH		HLSSR-GJ	
	Landsat-8 (LR)	Sentinel-2 (HR)	Landsat-8 (LR)	Sentinel-2 (HR)
Date	2020.04.13	2020.04.12	2020.04.15	2020.04.16
Spatial resolution	10 m	30 m	10 m	30 m
Area	16400 km^2		21670 km^2	
Bands	Blue, Green, Red, NIR, SWIR-1, SWIR-2			
Sample size	160×160	480×480	160×160	480×480
Number of training sets	2250	2250	2891	2891
Number of test sets	1100	1100	1225	1225

where RAG^k and RAG^{k-1} represent the kth RAG and $(k-1)$th RAG. s^l is the lth EAB. W is a convolutional layer whose role is to enhance the translational equivariance of transformer layers.

C. Feature Fusion Module

CNNs can fully mine the spatial and channel information in the receptive field, but lacks explicit modeling between channels [41]. On the contrary, transformers can make full use of long-range and global information through self-attention mechanism, but is limited in capturing local details due to the lack of spatial inductive biases [42]. Therefore, we designed the FFM to fuse deep features of different dimensions extracted by CNNs and transformers. The structure of FFM is shown in Fig. 3.

FFM can be formulated as

$$\mathrm{FFM} = W_1(\mathrm{CA}(\mathrm{RAG}) \otimes \mathrm{RAG}) + W_2(\mathrm{SA}(\mathrm{RCG}) \otimes \mathrm{RCG}) \tag{7}$$

where W_1 and W_2 represent the convolutional layer, which is used to automatically learn the weights of different features extracted by CNNs and transformers. CA and SA represent channel attention and spatial attention, respectively. They come from CBAM [43], and the formula is as follows:

$$\mathrm{CA} = \mathcal{S}\left(W_1\left(W_0\left(F^c_{\mathrm{avg}}\right)\right) + W_1\left(W_0\left(F^c_{\mathrm{max}}\right)\right)\right)$$

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$$\mathrm{SA} = \mathcal{S}\left(W\left(\mathrm{cat}\left(F_{\mathrm{avg}}^{h\times w}, F_{\mathrm{max}}^{h\times w}\right)\right)\right) \tag{8}$$

where F_{avg}^{c} and F_{max}^{c} represent the global average pooling and global maximum pooling based on width and height, respectively. $F_{\mathrm{avg}}^{h\times w}$ and $F_{\mathrm{max}}^{h\times w}$ represent the global average pooling and global maximum pooling based on channels, respectively. cat(·) is concatenation operation, and $S(\cdot)$ is sigmoid function.

D. Residual Convolution Group

Deep convolutional networks have been proven to improve SR performance, and residual structures can avoid network degradation [10], [13]. Therefore, we construct RCGs with continuous residual units to capture the details of LR while stabilizing model training. Given a residual unit, it can be expressed as

$$x^{l+1} = x^{l} + W_1\sigma\left(W_0\left(x^{l}\right)\right) \tag{9}$$

where σ represents the leaky ReLU activation function. We concatenate N residual units to form an RCG module. RCG can be expressed as

$$\begin{aligned} \mathrm{RCG}^{k} &= x^{N-1} + W_1\sigma\left(W_0\left(x^{N-1}\right)\right), \quad N \geq 1 \\ x^{0} &= \mathrm{RCG}^{k-1} \end{aligned} \tag{10}$$

where RCG^{k} is the kth RCG feature map.

E. Spectral Loss

Due to the different imaging mechanism, imaging time, and atmospheric conditions of different satellite sensors, the reflectance of LR and HR images is disturbed by noise, which causes the problem of domain shift. After normalizing for external radiation and atmospheric effects, the remote sensing ratio index can provide a more consistent spatial and temporal standard than surface reflectance [44]. Because the normalized index cancels out most of the multiplicative noise caused by illumination differences, cloud shadows, terrain variations, and atmospheric conditions. Building upon the aforementioned analysis and drawing upon the widely employed remote sensing ratio index, we introduce a comprehensive spectral loss function tailored for the RSISR task. This loss function serves to ensure the radiometric consistency between HR and LR images, while addressing the challenges posed by domain shift arising from imaging variations and noise. We use the normalized difference vegetation index (NDVI) [45] and the normalized difference water index (NDWI) [46] to construct the spectral loss, and the calculation formula is as follows:

$$\begin{aligned} L_{\mathrm{spe}} &= \frac{1}{h\times w\times 2} \parallel S^{\mathrm{HR}} - S^{\mathrm{SR}} \parallel_1 \\ S^{\mathrm{HR}} &= \mathrm{cat}\left(\mathrm{NDVI}^{\mathrm{HR}}, \mathrm{NDWI}^{\mathrm{HR}}\right) \\ S^{\mathrm{SR}} &= \mathrm{cat}\left(\mathrm{NDVI}^{\mathrm{SR}}, \mathrm{NDWI}^{\mathrm{SR}}\right) \\ \mathrm{NDVI} &= \frac{\mathrm{nir}-\mathrm{r}}{\mathrm{nir}+\mathrm{r}} \\ \mathrm{NDWI} &= \frac{\mathrm{g}-\mathrm{nir}}{\mathrm{g}+\mathrm{nir}} \end{aligned} \tag{11}$$

where $\mathcal{L}_{\mathrm{spe}}$ is the spectral loss. g, r, and nir represent the reflectance of green, red, and near-infrared (NIR) bands, respectively. Finally, the overall loss function can be defined as

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\mathrm{rec}} + \lambda\mathcal{L}_{\mathrm{spe}} \\ \mathcal{L}_{\mathrm{rec}} &= \frac{1}{h\times w\times c} \parallel I^{\mathrm{HR}} - \mathrm{I}^{\mathrm{SR}} \parallel_1 \end{aligned} \tag{12}$$

where $\mathcal{L}_{\mathrm{rec}}$ represents the image reconstruction loss, which can be calculated by L1 loss. I^{HR} and I^{SR} represent HR and SR image, respectively. λ is the weight of spectral loss.

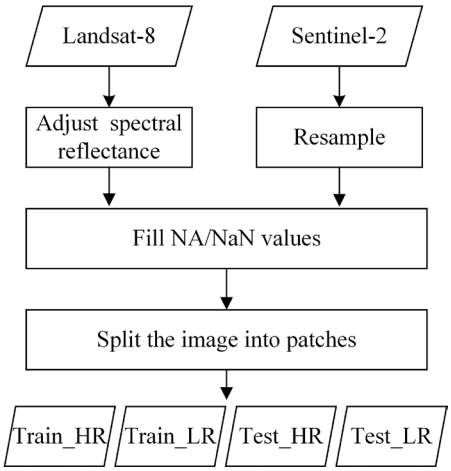


Fig. 5. Processing pipeline for generating the HLSSR-HH and HLSSR-GJ datasets.

IV. Experiments

A. Datasets

In order to address the lack of publicly available cross-sensor multispectral RSISR datasets, we have generated two harmonized datasets, namely HLSSR-HH and HLSSR-GJ, by aligning Landsat-8 and Sentinel-2 surface reflection data. The LR images were derived from atmospherically corrected surface reflectance images from Landsat-8 OLI/TIRS Level 2 Collection 2, while the HR images were obtained from orthorectified atmospherically corrected surface reflectance data of Sentinel-2 MSI Level-2A. Fig. 4 illustrates the geographic locations and sample examples of these two datasets. The HLSSR-HH dataset covers areas in Hubei Province and Henan Province in China, whereas the HLSSR-GJ dataset encompasses regions in Guangdong Province and Jiangxi Province in China. Landsat-8 and Sentinel-2 images were downloaded from Google Earth Engine. Further information regarding the two datasets is presented in Table I.

The processing pipeline for generating the HLSSR-HH and HLSSR-GJ datasets is depicted in Fig. 5. Initially, we employ the spectral band transformation functions introduced by Zhang et al. [47] to align the spectral reflectance of Landsat-8 images, ensuring greater spectral consistency between Landsat-8 OLI and Sentinel-2A MSI data. Subsequently, the Sentinel-2 image is resampled to precisely three times the size of the Landsat-8 image to establish a suitable spatial correspondence. Lastly, the images are divided into corresponding HR/LR pairs for further analysis and model training.

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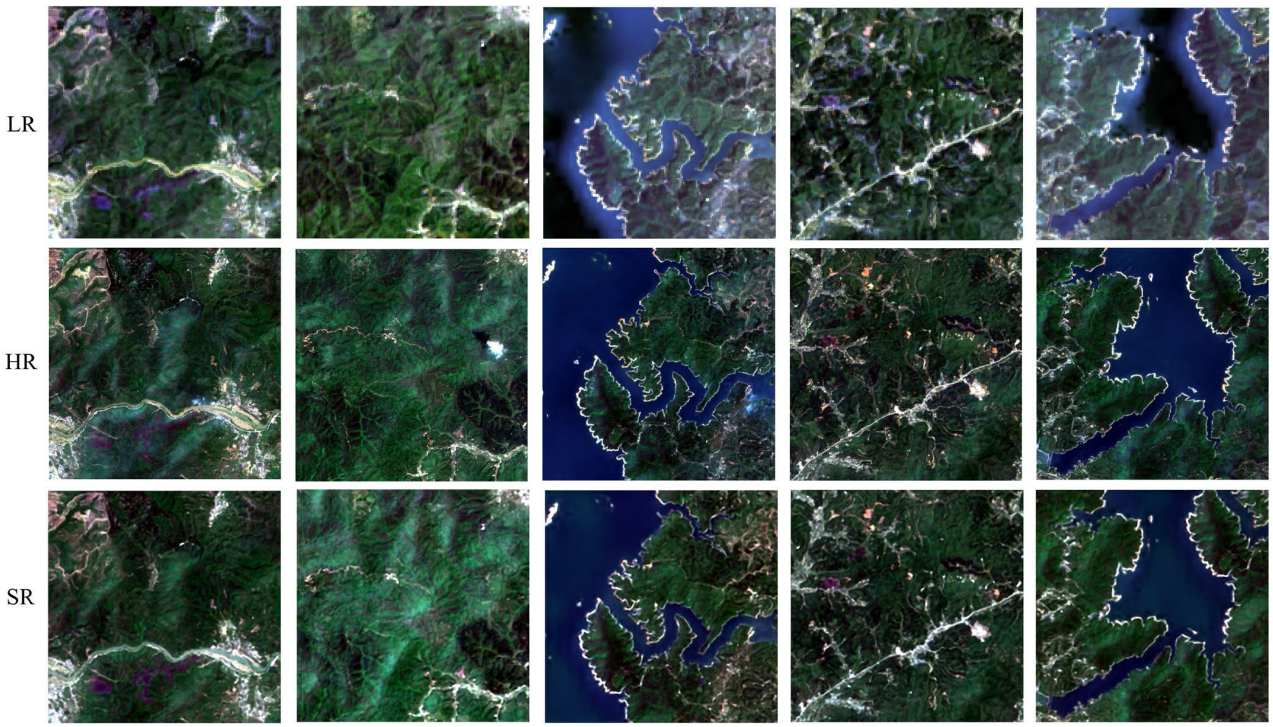


Fig. 6. Examples of SR results for the HLSSR-GJ dataset. The first row showcases the input Landsat-8 LR images, the second row displays the corresponding Sentinel-2 HR images, and the third row presents the SR results obtained by our ConvFormerSR model.

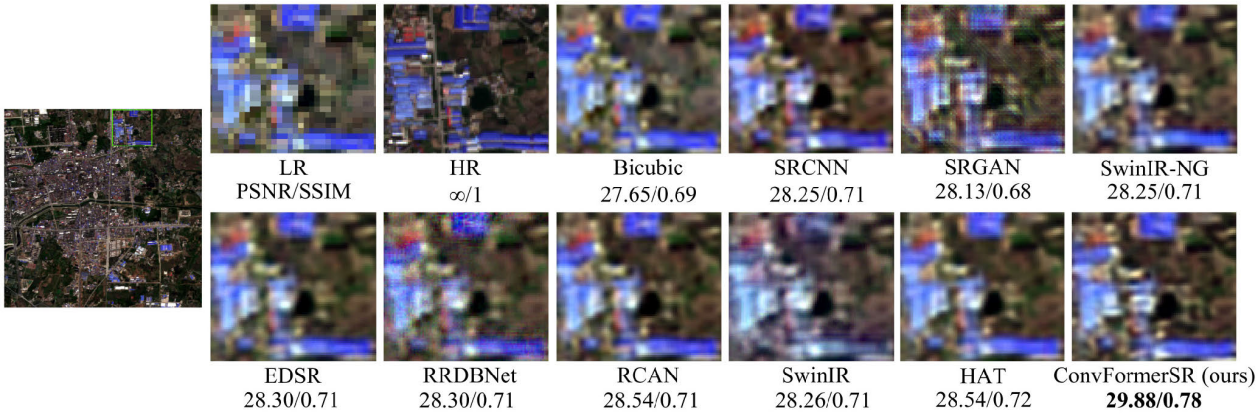


Fig. 7. Qualitative comparison of our model with other works on the HLSSR-HH dataset.

B. Implementation Details

1) Training Settings: We implemented our SR model using PyTorch and conducted all experiments on a high-performance computing platform equipped with Nvidia Tesla V100 16G GPUs. The model takes randomly cropped LR images of size 64 × 64 pixels as input, while the corresponding HR size is 192 × 192 pixels. To ensure computational efficiency, we selected four specific bands, namely blue, green, red, and NIR, for model training. The batch size was set to 10, and we randomly rotated HR/LR images for data augmentation. The learning rate was initially set to 1e−4 and reduced by a factor of 0.1 after 80 epochs. A total of 300 epochs were trained using Adam optimizer. Regarding the model's parameter settings, we configured the number of RAG, FFM, and RCG as 4. The reduction ratio α in GCAM was set to 3, and the weight of GCAM (γ) was set to 0.01. Furthermore, we employed 16 residual units in RCG. The parameter settings for transformer block in the RAG were consistent with SwinIR, with six EAB modules, a window size of 8, and six attention heads.

2) Evaluation Metrics: The peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [48] are commonly used evaluation metrics for SR. Given that the true value HR is x, and the reconstruction SR is y, the calculation formulas are as follows:

$$\begin{aligned} \mathrm{PSNR}(x,y) &= 10\log_{10}\frac{1}{\mathrm{MSE}(x,y)} \\ \mathrm{SSIM}(x,y) &= \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \end{aligned} \tag{13}$$

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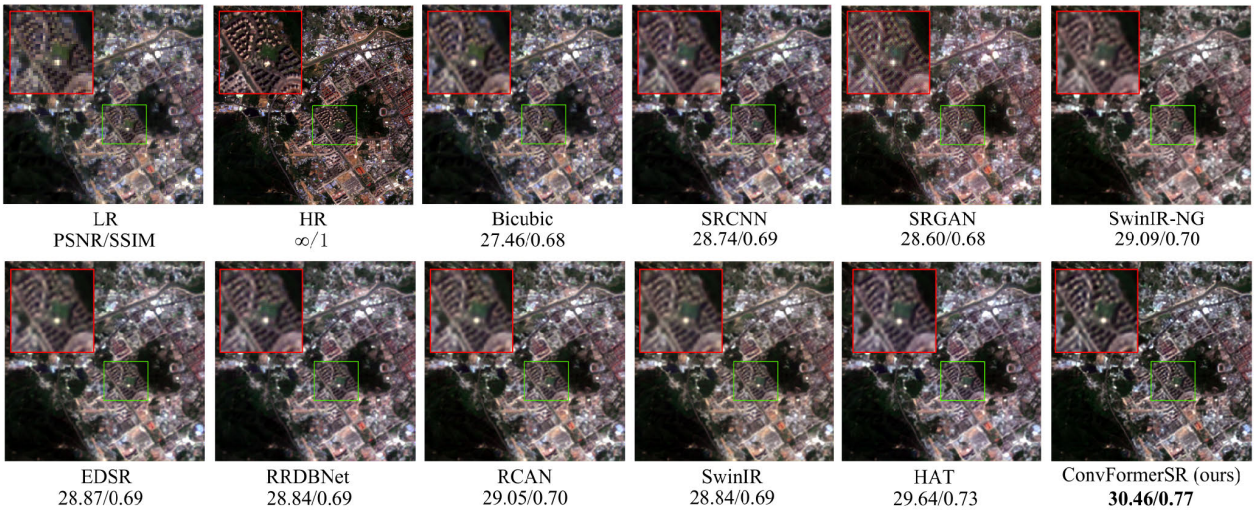


Fig. 8. Qualitative comparison of our model with other works on the HLSSR-GJ dataset.

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS FOR RSISR ON HLSSR-HH AND HLSSR-GJ DATASETS

Dataset	Method	PSNR (dB)	SSIM	RMSE	SAM	ERGAS
HLSSR-HH	Bicubic	32.99	0.8818	0.0226	0.1924	7.3961
	SRCNN	33.62	0.8902	0.0210	0.1812	6.7377
	SRGAN	33.49	0.8716	0.0214	0.1922	7.0916
	EDSR	33.74	0.8919	0.0208	0.1787	6.6194
	RRDBNet	33.61	0.8867	0.0211	0.1821	6.7643
	RCAN	34.22	0.8985	0.0197	0.1709	6.3447
	SwinIR	33.72	0.8901	0.0208	0.1797	6.6967
	HAT	33.97	0.8946	0.0203	0.1746	6.5329
	SwinIR-NG	33.87	0.8931	0.0205	0.1765	6.5462
	ConvFormerSR (ours)	**35.54**	**0.9214**	**0.0169**	**0.1463**	**5.5204**
HLSSR-GJ	Bicubic	30.23	0.8747	0.0311	0.2256	10.2059
	SRCNN	32.53	0.8867	0.0240	0.2130	8.4881
	SRGAN	33.01	0.8775	0.0228	0.2158	8.6123
	EDSR	33.23	0.8883	0.0222	0.2082	8.3069
	RRDBNet	33.21	0.8864	0.0223	0.2103	8.3753
	RCAN	33.80	0.8917	0.0208	0.2023	8.0987
	SwinIR	33.60	0.8894	0.0213	0.2050	8.2175
	HAT	33.95	0.8965	0.0205	0.1973	7.9566
	SwinIR-NG	33.64	0.8924	0.0212	0.2022	8.0810
	ConvFormerSR (ours)	**34.67**	**0.9089**	**0.0189**	**0.1827**	**7.2996**

where μ_x and σ_x represent the mean and variance of x, respectively. σ_{xy} represents the covariance of x and y. c_1 and c_2 are constants.

In addition, root-mean-square error (RMSE), spectral angle mapper (SAM) [49], and relative dimensionless global error in synthesis (ERGAS) [50] are also used as evaluation indicators, which are widely used to evaluate the quality of remote sensing image reconstruction. SAM represents the spectral similarity between the reconstructed image and the reference image, and it can be calculated as

$$\mathrm{SAM} = \cos^{-1} \frac{\sum xy}{\sqrt{\sum (x)^2 \sum (y)^2}}. \tag{14}$$

ERGAS represents the global quality of image fusion, which can be expressed as

$$\mathrm{ERGAS} = 100 \frac{h}{l} \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(\frac{\mathrm{RMSE}_i}{\mu_i} \right)^2} \tag{15}$$

where h and l represent the spatial resolution of the SR image and the original image, respectively, and N is the number of bands. The aforementioned metrics are computed for each individual band, after which their average value is employed as the overall reconstruction result. Higher values of PSNR and SSIM indicate better SR quality. Conversely, lower values of RMSE, SAM, and ERGAS suggest superior reconstruction quality.

C. Results

Taking the HLSSR-GJ dataset as an example, Fig. 6 shows five LR/HR image pairs from the test set and their corresponding SR results. It can be seen that our model can not only reconstruct the details of LR, but also correct the spectrum of LR to make it as consistent as possible with HR. We further conducted a comparative analysis with state-of-the-art methods, namely Bicubic, SRCNN [7], SRGAN [8], EDSR [10], RRDBNet [23], RCAN [13], SwinIR [16], HAT [26],

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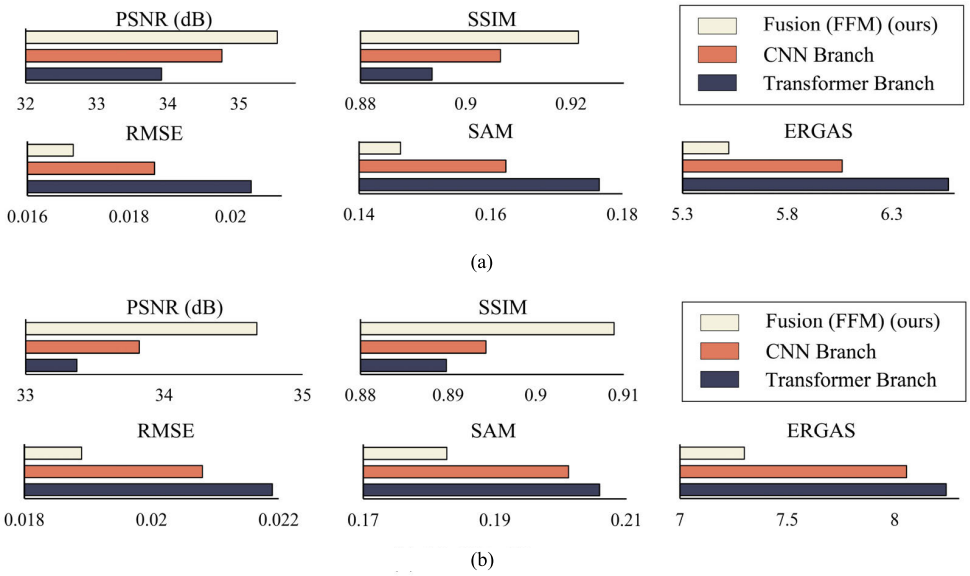


Fig. 9. Comparison of our model with different branch models on (a) HLSSR-HH and (b) HLSSR-GJ datasets.

TABLE III
PERFORMANCE COMPARISON OF MODELS WITH AND WITHOUT GCAM

	HLSSR-HH		HLSSR-GJ	
	w/o GCAM	w/ GCAM	w/o GCAM	w/ GCAM
PSNR (dB)	33.72	**35.54**	33.29	**34.67**
SSIM	0.8916	**0.9214**	0.8895	**0.9089**
RMSE	0.0208	**0.0169**	0.0221	**0.0189**
SAM	0.1795	**0.1463**	0.2071	**0.1827**
ERGAS	6.6684	**5.5204**	8.2393	**7.2996**

and SwinIR-NG [51]. Visual comparisons on the HLSSR-HH and HLSSR-GJ datasets are presented in Figs. 7 and 8, respectively. In Fig. 7, we focus on local zoom-in detail from different models' results. Fig. 8 showcases complete results, with the sub-image within the red rectangle representing an enlarged view of the green rectangle. It can be seen that our model is able to reconstruct clear details and sharper edges. In contrast, alternative models produce results that are either blurry or contain substantial noise and artifacts. For instance, the SR images generated by SRGAN and RRDBNet exhibit noticeable noise and artifacts, while the edges of the reconstructed images from SRCNN and EDSR appear excessively blurred.

Moreover, we conducted a comprehensive quantitative evaluation of our model on the two datasets. As depicted in Table II, our model outperforms other models in terms of PSNR, SSIM, RMSE, SAM, and ERGAS. Specifically, for HLSSR-HH dataset, our model achieves improvements ranging from 1.32 to 2.55 dB in PSNR and 0.02 to 0.05 in SSIM compared to other models. Our model also exhibits SAM values that are 0.02–0.05 lower than other models, and ERGAS values that are 0.82–1.88 lower. Similarly, for HLSSR-GJ dataset, our model achieves increases ranging from 0.72 to 4.44 dB in PSNR and 0.01 to 0.03 in SSIM, with SAM values that are 0.01–0.04 lower and ERGAS values that are lower by 0.66–2.91 compared to other models. Notably, our proposed model is the only one to achieve an SSIM metric exceeding 0.90 on both datasets. On the whole, the incorporation of attention mechanisms plays a crucial role in RSISR, and models based on attention mechanisms such as RCAN, HAT, SwinIR-NG, and ConvFormerSR demonstrate excellent performance.

V. DISCUSSION

A. Performance of Different Modules

1) Effect of FFM: To evaluate the effectiveness of FFM, we conducted experiments separately using the transformer branch and CNN branch. The results of these experiments are presented in Fig. 9. The comparison demonstrates that utilizing FFM yields superior performance compared to using a single branch model. ConvFormerSR exhibits higher values of PSNR and SSIM, as well as lower values of RMSE, SAM, and ERGAS. These findings indicate that our model effectively captures both local details and long-range dependencies. Additionally, we observed that the CNN branch outperforms the transformer branch on both datasets. This suggests that despite the limited ability of CNNs to model long-range dependencies, the residual-based deep CNNs still yields favorable results for pixel-intensive SR reconstruction tasks.

2) Effect of GCAM: To evaluate the effectiveness of GCAM, we removed the GCAM module and verified the accuracy of the test sets. The experimental results are summarized in Table III. By incorporating the GCAM module, we observed improvements in various metrics. On the HLSSR-HH dataset, the PSNR and SSIM were increased by 1.82 and 0.03, respectively, while the SAM and ERGAS were reduced by 0.03 and 1.15, respectively. Similarly, on the HLSSR-GJ dataset, the PSNR and SSIM were improved by 1.38 and 0.02, respectively, and the SAM and ERGAS were reduced by 0.02 and 0.94, respectively. These results clearly demonstrate that the addition of the GCAM module effectively enhances

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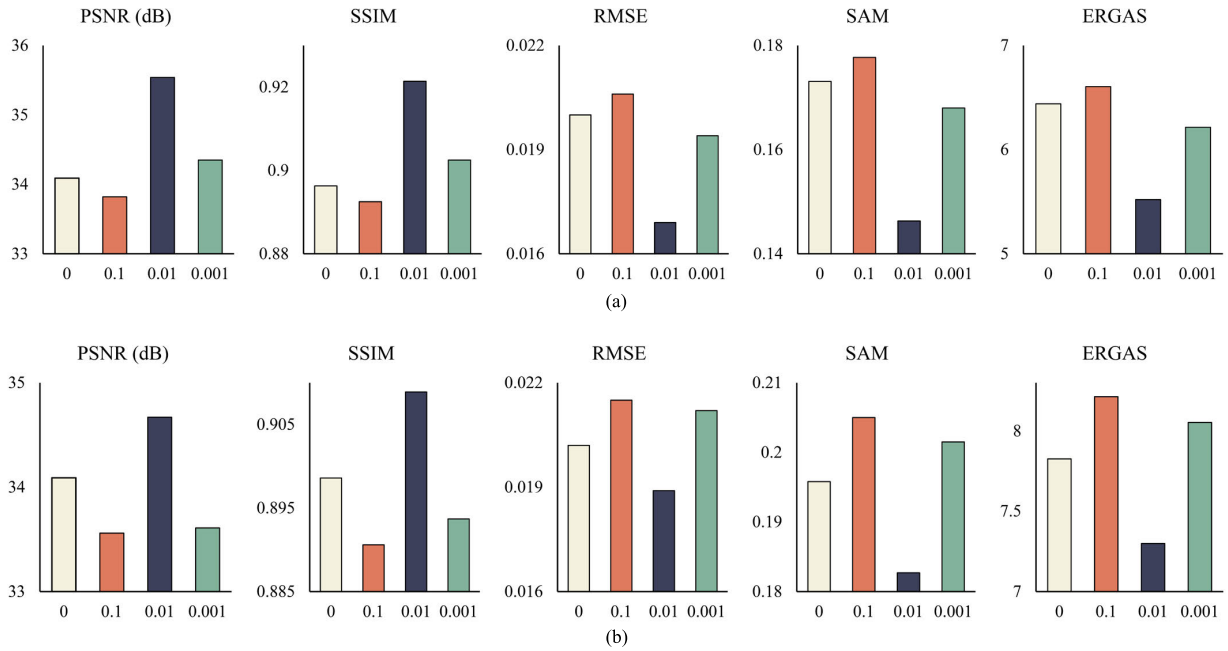


Fig. 10. Performance of different thresholds λ for (a) HLSSR-HH and (b) HLSSR-GJ datasets.

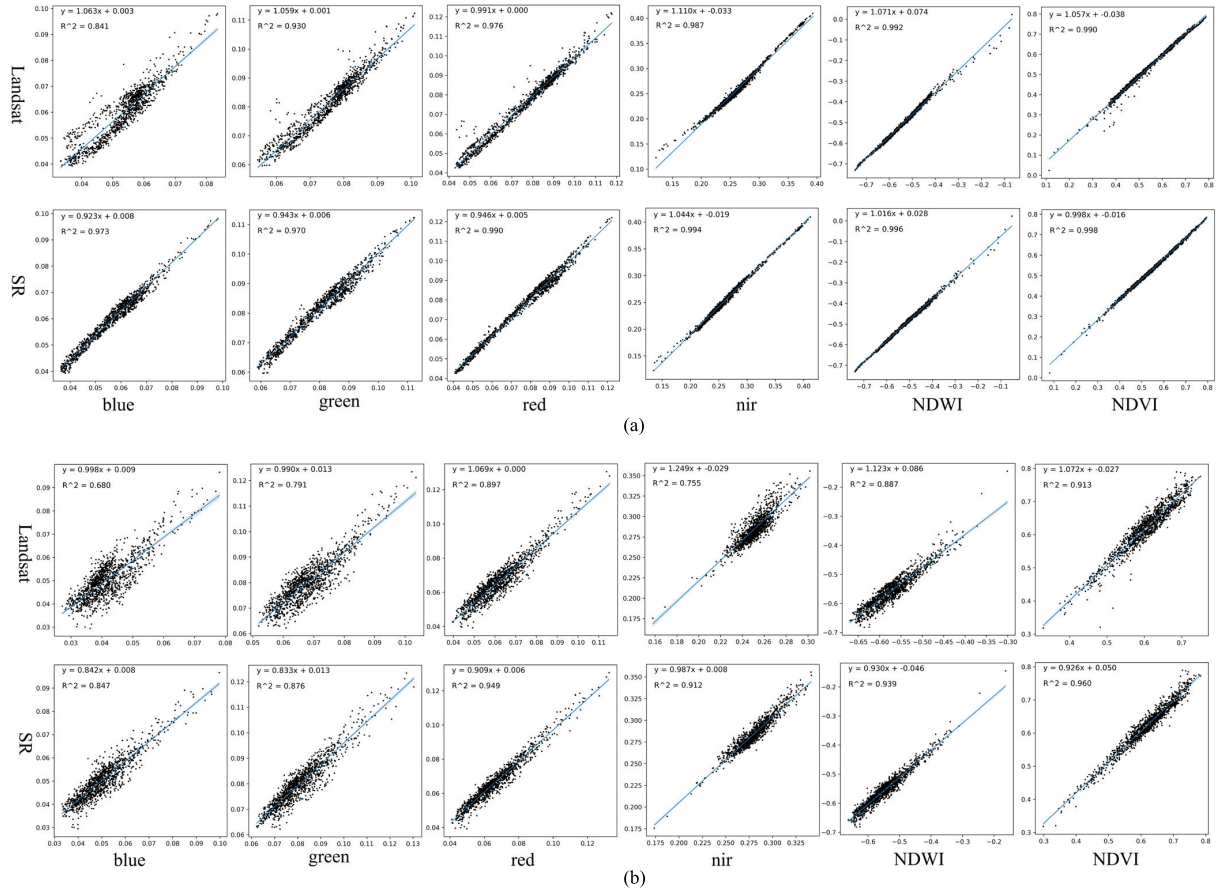


Fig. 11. Scatter diagram of spectral consistency assessment on (a) HLSSR-HH and (b) HLSSR-GJ datasets.

the global modeling capability of transformers and improves the SR performance of our model.

B. Analysis of Spectral Loss

In this section, we examine the influence of the parameter λ in our proposed spectral loss function, which plays a crucial role in cross-sensor RSISR. We conducted experiments with different values of λ, specifically 0, 0.1, 0.01, and 0.001, to investigate the impact on our model's performance. When λ is set to 0, the loss function reduces to the L1 loss. The detailed experimental results are presented in Fig. 10. Notably, when λ is set to 0.01, our model achieves the best performance,

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TABLE IV
COEFFICIENT OF DETERMINATION OF SR RESULTS OF DIFFERENT MODELS

			Blue	Green	Red	NIR	NDWI	NDVI
HLSSR-HH	LR	Landat-8	0.841	0.930	0.976	0.987	0.992	0.990
	SR	Bicubic	0.841	0.930	0.976	0.987	0.992	0.990
		SRCNN	0.932	0.951	0.988	0.993	0.987	0.993
		SRGAN	0.948	0.966	**0.992**	0.987	0.994	0.997
		EDSR	0.953	0.961	0.991	0.988	0.994	0.996
		RRDBNet	0.930	0.945	0.987	0.983	0.989	0.993
		RCAN	0.959	0.969	0.991	0.994	0.993	0.996
		SwinIR	0.951	0.960	0.989	0.987	0.993	0.995
		HAT	0.963	0.969	0.991	0.991	0.993	0.996
		SwinIR-NG	0.951	0.965	0.989	0.993	0.994	0.996
		ConvFormerSR	**0.973**	**0.970**	0.990	**0.994**	**0.996**	**0.998**
HLSSR-GJ	LR	Landat-8	0.680	0.791	0.897	0.755	0.887	0.913
	SR	Bicubic	0.680	0.791	0.897	0.755	0.887	0.913
		SRCNN	0.756	0.829	0.924	0.799	0.910	0.939
		SRGAN	0.763	0.826	0.924	0.850	0.911	0.937
		EDSR	0.770	0.842	0.932	0.838	0.919	0.945
		RRDBNet	0.762	0.838	0.932	0.831	0.913	0.941
		RCAN	0.762	0.828	0.930	0.902	0.925	0.951
		SwinIR	0.780	0.836	0.929	0.898	0.923	0.947
		HAT	0.790	0.851	0.940	0.911	0.926	0.949
		SwinIR-NG	0.788	0.851	0.940	0.891	0.926	0.953
		ConvFormerSR	**0.847**	**0.876**	**0.949**	**0.912**	**0.939**	**0.960**

TABLE V
COMPARISONS OF MODEL COMPLEXITY AND EFFICIENCY

Model	#Params (M)	FLOPs (G)	FPS
Bicubic	-	-	5335.41
SRCNN	0.03	6.15	152.61
SRGAN	6.00	26.71	33.11
EDSR	43.68	1119.72	4.33
RRDBNet	16.66	441.78	4.11
RCAN	15.63	397.28	3.61
SwinIR	16.57	424.84	1.99
HAT	26.01	668.88	0.18
SwinIR-NG	14.66	309.77	1.16
ConvFormerSR	15.56	398.93	2.13

and this trend remains consistent across both datasets. This indicates that the spectral loss effectively mitigates radiation differences arising from varying illumination conditions, cloud shadows, terrain variations, and atmospheric conditions. However, the weight assigned to the spectral loss is a critical factor. When the weight is too large ($\lambda = 0.1$), the inclusion of spectral loss diminishes the overall accuracy of our model. This can be attributed to the fact that the spectral loss, which employs radiation ratios, cannot fully capture the genuine surface reflectance. Conversely, if the weight is too small ($\lambda = 0.001$), the effectiveness of the spectral loss is weakened. Hence, an appropriate weight for the spectral loss proves instrumental in addressing the domain shift issue encountered in cross-sensor RSISR.

C. Spectral Consistency Assessment

To assess the spectral consistency after SR and to ensure that the RSISR reconstruction preserves the original surface radiation values, we conducted an analysis using the surface reflectance of Sentinel-2 images from the test sets as the dependent variable and the Landsat-8 and SR images as independent variables. Scatter plots were generated, and linear fitting was performed to evaluate the relationship between the variables. Additionally, we calculated the NDWI and NDVI. The comparison between ConvFormerSR results and Landsat images is shown in Fig. 11, and the coefficients of determination (R^2) of all model reconstruction results in different bands are shown in Table IV. Observing the figures and table, it is evident that the normalized indices (NDWI and NDVI) exhibit a higher correlation compared to the surface reflectance values (blue, green, red, and NIR). Notably, the R^2 of all bands in the SR results is significantly enhanced compared to Landsat-8 LR images, and the SR surface reflectance demonstrates greater consistency with Sentinel-2 images. Furthermore, our proposed model has the highest R^2 in all bands except the red band of HLSSR-HH dataset. These findings illustrate that our proposed framework not only improves the spatial resolution of the imagery through SR, but also ensures effective radiometric calibration, thereby preserving the desired radiometric properties.

D. Complexity and Efficiency Evaluation

In order to quantitatively evaluate the complexity and computational efficiency of our proposed model, we calculated the number of training parameters (#Params), floating-point operations (FLOPs), and frames per second (FPS) of different models. The experimental results are shown in Table V. The bicubic interpolation technique inherently circumvents the necessity for parameter optimization, thereby manifesting the swiftest inference speed within the scope of our evaluation.

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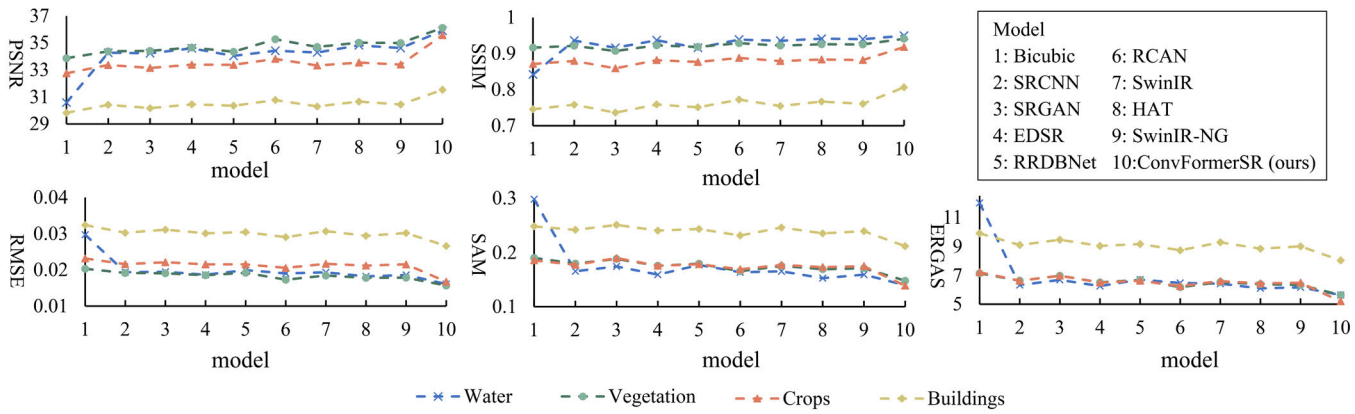


Fig. 12. Performance of different land-cover categories on HLSSR-HH dataset.

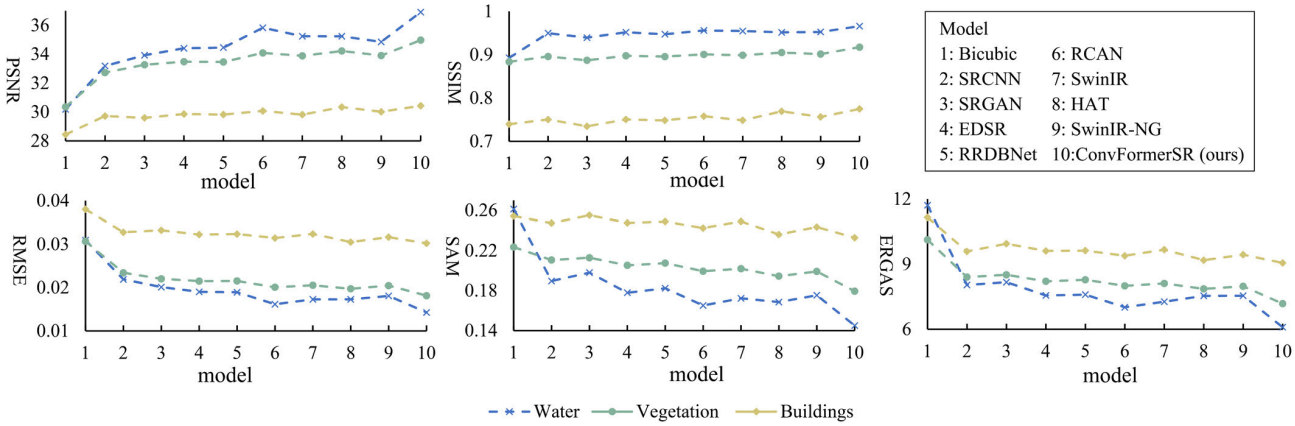


Fig. 13. Performance of different land-cover categories on HLSSR-GJ dataset.

Conversely, EDSR model has the largest number of parameters and FLOPs due to its deep residual structure. The HAT model, incorporating an enlarged window size and introducing supplementary cross-attention blocks, engenders a network of elevated complexity, which leads to a slow inference speed. Combining Tables II and V, it can be found that the complexity and inference speed of our proposed model are comparable to most models, but the performance of our model is better, which fully demonstrates the potential of our model to be applied in practice.

E. Effect of Land Cover on SR

To explore the impact of land cover on model performance, we undertook a systematic assessment of model accuracy across distinct land-cover categories. The land cover for both datasets emanated from the Dynamic World [52], which provides a pixel-level classification map corresponding to each Sentinel-2 image. The experimental results are shown in Figs. 12 and 13. In the HLSSR-HH test set, the land-cover types mainly include water bodies, vegetation (trees, grass, and shrubs), crops, and buildings. The SR reconstruction accuracy is highest for vegetation and water, while the SR accuracy for buildings is the lowest. Apart from bicubic interpolation, the SR performance of all deep learning models exhibits consistency across different land-cover categories. Furthermore, across all categories, the performance of our proposed model is significantly superior to that of other models. In the HLSSR-GJ test set, the primary ground cover consists of water, vegetation, and buildings. The SR accuracy is highest for water, whereas the accuracy for buildings is the lowest, consistent with the performance observed in HLSSR-HH dataset. In summary, water and vegetation surfaces exhibit a relatively smooth texture characterized by predominant low-frequency components, which results in a higher SR accuracy. On the other hand, the surface of buildings is abundant in intricate details and contains a substantial amount of high-frequency information. The recovery and reconstruction of such high-frequency information pose greater challenges, leading to a relatively lower level of SR accuracy.

F. Performance of Different Scale Factors

As the SR upscaling factor increases, the reconstructive capacity of the model diminishes. To explore the adaptability of our model to different scale factors, we conducted SR experiments simulating various upscaling ratios. Specifically, we downsampled the LR images while maintaining the HR images unchanged, resulting in HR image dimensions that are four times, five times, and six times larger than their corresponding LR counterparts. The experimental outcomes are presented in Table VI. It is evident that with the increase

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TABLE VI
PERFORMANCE OF DIFFERENT SCALE FACTORS

Dataset	Scale	PSNR (dB)	SSIM	RMSE	SAM	ERGAS
HLSSR-HH	×3	35.54	0.9214	0.0169	0.1463	5.5204
	×4	33.39	0.8850	0.0217	0.1862	5.2438
	×5	32.69	0.8739	0.0235	0.1996	4.4465
	×6	32.33	0.8691	0.0244	0.2053	3.7876
HLSSR-GJ	×3	34.67	0.9089	0.0189	0.1827	7.2996
	×4	34.14	0.8967	0.0201	0.1961	5.8467
	×5	33.04	0.8789	0.0227	0.2180	5.2010
	×6	32.99	0.8760	0.0228	0.2203	4.3863

in scale factors, the model's performance gradually deteriorates, owing to the heightened necessity to reconstruct finer details. Moreover, it should be noted that the upscaling factor influences the computation of the ERGAS metric. Therefore, ERGAS is not suited for comparisons between distinct SR scales and ERGAS values exhibit anomalous patterns in such scenarios.

G. Limitations and Future Research

First, we restricted our experiments to performing SR on only four bands considering computational efficiency. However, in practical scenarios, multispectral or hyperspectral images typically consist of a larger number of bands. To accommodate a greater number of input bands, we plan to simplify the model parameters accordingly. Furthermore, cross-sensor RSISR relies on the availability of LR/HR image pairs captured at the same time and location. However, it is often challenging to obtain suitable LR/HR pairs in certain areas. Consequently, enhancing the generalization capability of our model across a wider range or achieving cross-sensor blind SR becomes a primary focus of our future research endeavors. We aim to develop methods that can effectively address these challenges and enable robust and accurate SR even in scenarios where LR/HR pairs are not readily available.

VI. CONCLUSION

This article presents a novel approach, ConvFormerSR, specifically designed for cross-sensor RSISR. Our method aims to overcome the limitations of existing SR networks and effectively addresses the domain shift issues caused by different sensors. The proposed ConvFormerSR combines the strengths of CNNs and transformers to leverage their complementary abilities. By integrating the RAG module with the added GCAM, we effectively capture global relationships and high-order spatial interactions, thereby enhancing the performance of the traditional transformer structure. Our experimental findings demonstrate that a single transformer alone does not outperform CNN-based SR models, highlighting the importance of fusing CNNs and transformers in achieving superior results. To comprehensively capture local and global information from different branches, we introduce FFM to fuse the locality and translational equivariance of CNNs with the long-range dependency modeling capabilities of transformers. Additionally, to address the domain shift challenges caused by various factors such as illumination variations, cloud shadows, terrain changes, and atmospheric conditions, we propose a comprehensive spectral loss. Experimental results on two datasets validate the effectiveness of our proposed method in significantly improving the spatial resolution of Landsat-8 images while maintaining good spectral consistency in the SR results. In future work, we aim to explore the development of a blind SR model that can achieve cross-sensor SR even when the HR/LR pairs are inconsistent in terms of time and space.

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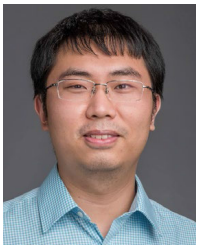
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